

NON-CONFORMANCE REPORT (NCR) & CAPA RECORD

AS9100D §8.7 Control of Nonconforming Outputs | IPC-A-610 Class 3 / J-STD-001

NCR #: NCR-2026-089 | Status: OPEN | Archived: 2026-09-24 12:40:11

1. HARDWARE IDENTIFICATION & DISCREPANCY PROFILE

NCR Number:	NCR-2026-089	Severity Class:	Critical (Class 3)
Assembly Part #:	PCA-8840-MCU (Rev D)	Current Status:	Open
Board Serial (S/N):	01051478	Containment Date:	2026-08-28
Standard Clause:	AS9100D §8.7 / IPC-A-610 Class 3	Assigned Owner:	Manufacturing Engineering

2. DEFECT DESCRIPTION & OBJECTIVE NON-CONFORMANCE EVIDENCE

Solder bridging observed on U4 fine-pitch QFN-48 pins during post-reflow automated optical inspection (AOI).

3. ROOT CAUSE ANALYSIS [METHODOLOGY: 5-WHY]

Solder paste volume slightly high on stencil aperture; reflow profile peak temp 242°C with minimal soak.

4. IMMEDIATE CONTAINMENT & CORRECTIVE / PREVENTATIVE ACTIONS (CAPA)

Reduce stencil aperture by 12% on QFN pads; retrain AOI operator on secondary visual gate.

5. AUDIT TRAIL, RECORD REVISIONS & MODIFICATION TIMESTAMPS

Date & Time (UTC/Local)	Author / Role	Change Details & Lifecycle Status
2026-09-24 12:40:11	Lead Quality Engineer	NCR details updated and verified.
2026-08-28 09:15:00	AOI Lead Inspector	Initial NCR logged from AOI discrepancy gate.
2026-08-28 14:32:15	Lead Quality Engineer	Updated containment plan and initiated 5-Why root cause

6. QUALITY ENGINEERING VERIFICATION & CONCURRENCE

Reviewed & Verified By: Lead Quality Engineer Sign-off Timestamp: 2026-09-24 12:40:11
QMS Authorization: AS9100D §8.7 / IPC-A-610 Certified Delegated Quality Authority Decision: Disposition verified & closed in compliance matrix.